

# Yoshitaka Inoue

Ph.D. Candidate in Computer Science · NIH Predoctoral Fellow

University of Minnesota, Twin Cities · National Institutes of Health

Bethesda, Maryland

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## Research Interests

Machine learning for biological perturbations and therapeutic response; single-cell and functional genomics; intervention-aware representation learning; graph-based biomedical reasoning; LLM-based scientific reasoning.

## Research and Professional Experience

**Pre-doctoral Fellow, National Library of Medicine / National Cancer Institute, NIH**

2024–present

*Bethesda, Maryland*

Developing machine learning methods for perturbation modeling and therapeutic response prediction, including treatment-conditioned latent-state transition models, graph-based drug-response models, and biomedical reasoning systems across single-cell, pharmacogenomic, high-throughput screening, cancer cell-line, and patient datasets.

**Google Summer of Code – Contributor (2022); Mentor (2023–2024)**

2022–2024

Developed a drug-response prediction pipeline and mentored projects on PyTorch Geometric datasets and multi-tool LLM aggregation for biomedical analysis.

**Machine Learning Engineer, FreakOut, Inc.**

2019–2021

*Tokyo, Japan*

Applied machine learning to click-through-rate prediction.

## Education

**Ph.D. Candidate in Computer Science and Engineering, University of Minnesota, Twin Cities**

2021–present

*Advisor: Prof. Rui Kuang; Co-advisor: Dr. Augustin Luna (NIH)*

Research on treatment-conditioned perturbation modeling, therapeutic response prediction, single-cell and functional genomics. Expected graduation: September 2027.

**M.Eng. in Information Science, Nara Institute of Science and Technology**

2017–2019

*Advisor: Prof. Shigehiko Kanaya*

Computational prediction and annotation of small molecules from tandem mass spectrometry (MS/MS) data in collaboration with the UC Davis Genome Center and DataLab.

**Visiting Scholar, University of California, Davis – Genome Center and DataLab**

Summer 2018

*Advisors: Prof. Oliver Fiehn and Tobias Kind*

Collaborative research on MS/MS-based compound prediction and metabolomics data analysis.

**B.Sc. in Biological Science, Kumamoto University**

2012–2017

*Advisor: Prof. Susumu Takio*

Experimental analysis of seagrass metabolism using molecular biology methods, including Northern blotting.

## Publications

1. Inoue, Y., Jeong, M., Hero, A., Kuang, R., & Luna, A. **PerturbRx: Learning Treatment-Conditioned Latent Transitions for Patient Drug Response Prediction.** *arXiv*, 2026; submitted to ICLR 2027.
2. Inoue, Y., Song, T., Wang, X., Luna, A., & Fu, T. **DrugAgent: Reliable Multi-Agent Aggregation under Conflicting Biomedical Evidence.** *Scientific Reports*, 2026, under revision.
3. Inoue, Y., Lee, H., Fu, T., Kuang, R., & Luna, A. **drGT: Attention-Guided Gene Assessment of Drug Response Utilizing a Drug-Cell-Gene Heterogeneous Network.** *BMC Bioinformatics*, 2026.
4. Inoue, Y., Hao, N., Lu, Y., Fu, T., & Luna, A. **Drug Discovery in the Era of Artificial Intelligence: From Target Identification to Clinical Trials.** *Health Data Science*, 2026, accepted.

- Hoshi-Kadonosawa, Y., Reinhold, W. C., Taniyama, D., Inoue, Y., et al. **TRPM4 Expression as a Predictive Biomarker and a Mechanistic Driver of Acetalax Activity in Prostate Cancer: Preclinical Efficacy Studies.** *Molecular Cancer Therapeutics*, 2026.
- Elloumi, F., Reinhold, W. C., Varma, S., Wang, Y., Kinali, M., Arakawa, Y., Inoue, Y., et al. **CellMiner cross-database (CellMiner-CDB) version 2.2 for explorations of patient-derived cancer cell line pharmacogenomics.** *Nucleic Acids Research*, 2026.
- Taniyama, D., Inoue, Y., Elloumi, F., Kim, Y. S., et al. **Spontaneous dynamic gene expression of Schlafen 11 (SLFN11) demonstrated by single-cell RNA sequencing in clonal cancer cell lines.** *Oncogene*, 2026, submitted.
- Inoue, Y., Kulman, E., & Kuang, R. **BiGCN: Leveraging Cell and Gene Similarities for Single-cell Transcriptome Imputation with Bi-Graph Convolutional Networks.** *bioRxiv*, 2024.

## Conference and Workshop Papers

- Inoue, Y., Song, T., Wang, X., Luna, A., & Fu, T. **DrugAgent: Reliable Multi-Agent Aggregation under Conflicting Biomedical Evidence.** *ICML Workshop on Foundation Models for the Life Sciences (FM4LS)*, 2026.
- Inoue, Y., Fu, T., & Luna, A. **GraphPINE: Graph Importance Propagation for Interpretable Drug Response Prediction.** *ICLR Workshop on Machine Learning for Genomics Explorations (MLGenX)*, 2025.
- Inoue, Y., Song, T., Wang, X., Luna, A., & Fu, T. **DrugAgent: Explainable Drug Repurposing Agent with Large Language Model-based Reasoning.** *ICLR Workshop on Machine Learning for Genomics Explorations (MLGenX)*, 2025.

## Research Software

### OncoGraph

Open oncology knowledge graph and retrieval benchmark

Integrates drugs, genes, diseases, clinical trials, Gene Ontology terms, and provenance-linked biomedical evidence. [github.com/inoue0426/OncoGraph](https://github.com/inoue0426/OncoGraph)

## Selected Presentations

### International Conference on Intelligent Systems for Molecular Biology (ISMB)

2026

Washington, DC

Two accepted oral presentations: drGT in NetBio and DrugAgent in Text Mining; both also presented at the Student Council Symposium.

### 26th Annual CCR-FYI Colloquium

2026

Rockville, MD

drGT oral presentation; selected as one of four Outstanding Postgraduate Fellow Finalists.

### NIH Artificial Intelligence Symposium

2025

Rockville, MD

drGT: attention-guided gene assessment for drug-response modeling; selected among 5 of 78 listed abstracts.

## Honors and Fellowships

- **NIH Pre-Doctoral Research Fellowship**, 2024–present.
- **Outstanding Postgraduate Fellow Finalist**, 26th Annual CCR-FYI Colloquium, 2026; selected as 1 of 4 finalists.
- **Research Visit Scholarships**, NAIST and JASSO support for UC Davis research visits, 2018–2019.

## Academic Service

**Journal reviewing.** BMJ Digital Health & AI (Expert Peer Reviewer, 2026–2027); IEEE/ACM Transactions on Computational Biology and Bioinformatics (2025–2026); Nature Computational Science (co-reviewer, 2026).

**Conference reviewing.** ACM BCB (2023); ICML FM4LS (2026); ICML GenBio (2026); IEEE BIBM (2026); ACM SIGSPATIAL (2026).

## Technical Skills

**ML / AI:** perturbation and transition modeling; graph neural networks; representation learning; LLM-based agents; interpretable machine learning.

**Computational biology:** pharmacogenomics; functional genomics; biological networks; single-cell RNA-seq; therapeutic response modeling; multi-omics analysis.

**Programming:** Python; PyTorch; PyTorch Geometric; scikit-learn; pandas; NumPy.

**Infrastructure:** Git; Linux; Jupyter; Slurm; GPU computing; HPC environments; LangChain.